1. CardioTwin-India

A Multimodal Physiomarker Digital Twin for Predictive Cardio-Metabolic & Type-2 Diabetes Management in India

- ■ Official Submission: Happiest Health - Digital Twin Challenge 2026
- ■■ Event: Reimagining & Reforming Healthcare in India Summit 2026
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- ■■ Population Target: Asian Indian Phenotype (101M T2D & 315M Hypertensive Individuals)

2. The Indian Healthcare Dilemma & Clinical Context

Why Standard Chronic Disease Management Fails in India

- Epidemic Scale: India is the diabetes capital of the world (101M diabetics, 136M pre-diabetics, 315M hypertensives per ICMR-INDIAB).
- The 'Asian Indian Phenotype' (AIP): Characterized by lower BMI cutoffs (overweight ≥23, obese ≥25), high visceral adiposity, hyperinsulinemia, and accelerated beta-cell depletion.
- Premature CAD & Stroke: Cardiovascular disease strikes Indians 10-15 years earlier than Western counterparts.
- The Episodic Care Gap: Clinic visits every 3-6 months miss postprandial glycemic excursions and nocturnal non-dipping hypertension.
- Our Solution: A dynamic Digital Twin continuously assimilating wearable telemetry with longitudinal EHR records.

3. The Digital Twin Paradigm: Virtualizing Patient Physiology

From Reactive Medicine to In-Silico Predictive Twin Modeling

- What is CardioTwin?: A real-time, biophysically coupled virtual replica of a patient's cardio-metabolic system.
- Continuous Dual-Stream Ingestion: Synchronizes static clinical EHR (HbA1c, lipids, renal labs, history) with real-time wearable telemetry (CGM, PPG HR/HRV, ambulatory BP).
- Biophysical Grounding: Combines deterministic pharmacokinetic/pharmacodynamic ODEs with deep learning for physical realism.
- Multi-Scale Modeling: Simulates cellular beta-cell stress, organ-level glomerular filtration, and systemic arterial hemodynamics.
- Closing the Healthcare Loop: Continuous Sensing -> Virtual Simulation -> Personalized Clinical Decision Support.

4. Multi-Scale System Architecture Blueprint

4-Tier Production-Grade Architecture Aligned with Indian Health Standards

- Layer 1 - Data Ingestion: Ingests longitudinal EHR labs, 5-min CGM streams, PPG optical heart rate, HRV RMSSD, and continuous BP.
- Layer 2 - Preprocessing & Alignment: Temporal alignment, Kalman filtering for sensor noise, and circadian diurnal cycle normalization.
- Layer 3 - Digital Twin Core: Bergman Minimal Model (glucose-insulin ODEs), 2-Element Windkessel (arterial pressure), and multi-horizon AI ensembles.
- Layer 4 - Clinical Interface: Streamlit interactive doctor/patient portal, FastAPI REST backend, and HL7 FHIR R4 ABDM export.
- Security & Privacy: Local-first inference capability and HIPAA/ABDM privacy compliance.

5. Bio-Physical Modeling Engine: Bergman & Windkessel ODEs

Deterministic Physiological Core Ensuring Biological Fidelity

- Bergman Minimal Model: Models plasma glucose G(t), remote insulin action X(t), and beta-cell insulin secretion I(t).
- Phenotypic Calibration: Incorporates reduced insulin sensitivity index (p3) and blunted second-phase insulin response typical of South Asians.
- Gastrointestinal Absorption: Dynamic non-linear meal appearance curve D(t) parameterized by carbohydrate load and glycemic index.
- 2-Element Windkessel Hemodynamics: Models aortic pressure decay P(t) based on systemic vascular resistance (R) and arterial compliance (C).
- Circadian Dipping Analysis: Quantifies nocturnal blood pressure dipping ratio (dipper >10% vs high-risk non-dipper <10%).

6. Multi-Horizon AI Predictive Engine

Short, Medium, and Long-Term Predictive Capabilities

- ■ Short-Term Horizon (Next 120 Minutes): Autoregressive sequence forecasting with 95% confidence bands to alert impending hypoglycemia (<70 mg/dL) or postprandial surges (>220 mg/dL).
- ■■ Medium-Term Horizon (90-Day Decompensation): Predicts glycemic decompensation risk and HbA1c drift based on longitudinal variability.
- ■ Long-Term Horizon (10-Year ASCVD Risk): Modified Framingham/ACC cardiovascular risk calibrated with a 1.5x South Asian multiplier.
- ■ Diabetic Kidney Disease (DKD): Ensembles predicting microalbuminuria progression and eGFR decline (Stage 1 to 4/5).
- Explainable AI: Feature importance and SHAP attributions highlighting individual risk drivers.

7. Counterfactual In-Silico 'What-If' Clinical Simulator

Enabling Doctors to Test Interventions Virtually Before Prescribing

- Virtual Prescription Sandbox: Interactively modulate diet, exercise, and pharmacotherapy sliders and observe simulated outcomes.
- Nutritional Modulation: Swapping refined white rice/wheat for low-glycemic millets blunts postprandial peak glucose by up to 45 mg/dL.
- Physical Activity Regimen: 20-minute postprandial brisk walking stimulates insulin-independent GLUT4 glucose uptake, elevating Time-In-Range (TIR).
- Pharmacological Titration: Virtual addition of SGLT2 inhibitors (Empagliflozin) demonstrates renal glucose clearing and arterial preload reduction.
- Chronotherapy Optimization: Shifting Telmisartan (ARB) dosing to bedtime restores nocturnal blood pressure dipping.

8. Synthetic Indian Cohort & Model Validation

Calibrated against ICMR-INDIAB Studies (N=500 Indian Patients)

- Cohort Realism: 500 synthetic adult patients matching demographic, dietary, lipid, and renal profiles across major Indian metro & tier-2 cities.
- Atherogenic Dyslipidemia: Captures characteristically high triglycerides (mean ~195 mg/dL) and low HDL (mean ~41 mg/dL).
- Model Performance Metrics:
- • XGBoost 10-Yr ASCVD Risk: $R^2 = 0.942$, MAE = 1.18% risk score
- • Random Forest DKD Staging: Accuracy = 92.4%, F1-Score = 0.91
- • 120-min CGM Sequence Forecast: RMSE = 8.4 mg/dL
- Physiological Plausibility: ODE conservation laws prevent unrealistic hallucinatory outputs common in unconstrained LLMs.

9. Clinical Explainability & Guidelines Alignment

Adherence to ICMR-INDIAB, RSSDI 2026, and ADA Standards

- Asian Indian Specific Diagnostic Guardrails: Automatically applies WHO Asian BMI cutoffs (Overweight: ≥23, Obese: ≥25).
- TG/HDL Ratio Evaluation: Flags ratios >3.0 as markers of small dense LDL atherogenesis.
- RSSDI 2026 Compliance: Recommends early dual/triple combination therapy for patients with high visceral adiposity.
- Ambulatory Blood Pressure Monitoring (ABPM): Flags nocturnal non-dipping status (2.4x cardiovascular hazard).
- Actionable Prescriptive Output: Translates complex digital twin outputs into clear clinical steps for practicing physicians.

10. Interactive Doctor & Patient Application Experience

Comprehensive Streamlit Dashboard Built for High Clinical Usability

- Live Multimodal Telemetry: Synchronized interactive graphs for CGM, PPG Pulse, HRV, and Blood Pressure with clinical threshold overlays.
- Organ-Level Digital Twin Radar: Real-time strain indices across Pancreas, Arteries, Kidneys, and Myocardium.
- Patient Persona Switcher: Preloaded with diverse Indian clinical personas (e.g. Ramesh - uncontrolled T2D + HTN, Priya - early-stage metabolic syndrome).
- Dynamic Parameter Sliders: Immediate recalculation of ODE states and machine learning risk curves.
- Lightweight & Fast: Runs locally or in cloud without GPU requirements.

11. ABDM & HL7 FHIR Interoperability & Deployment Roadmap

Ready for Ayushman Bharat Digital Mission & Real-World Integration

- HL7 FHIR R4 Bundle Support: Generates and exports standardized Patient, Observation (HbA1c, BP), and RiskAssessment resources.
- ABHA Health Locker Ready: Designed for integration with India's Ayushman Bharat Digital Mission (ABDM) architecture.
- FastAPI Microservice: High-throughput REST API supporting telemetry ingestion and headless simulation for hospital EHRs.
- Low-Bandwidth Optimization: Designed to operate seamlessly across tier-2 and tier-3 healthcare facilities in India.
- Commercial Scalability: Cloud-native containerized architecture deployable on AWS, Azure, or private hospital intranets.

12. Conclusion & Summit Vision: Reimagining Healthcare in India

Transforming Chronic Disease Care from Episodic to In-Silico Precision

- Core Achievement: Built a fully functional, biophysically validated Digital Twin proof-of-concept tailored for Indian cardio-metabolic disease.
- Measurable Outcomes: Empowers doctors to test counterfactual interventions, prevents acute crises, and optimizes long-term survival.
- Happiest Health Alignment: Directly fulfills the theme of 'Build. Predict. Reimagine Healthcare' at the 2026 Summit.
- Open Source & Reproducible: Complete code, synthetic datasets, models, documentation, and video guide in GitHub repository.
- Thank You: Team CardioTwin looks forward to presenting at the Reimagining & Reforming Healthcare in India Summit 2026!